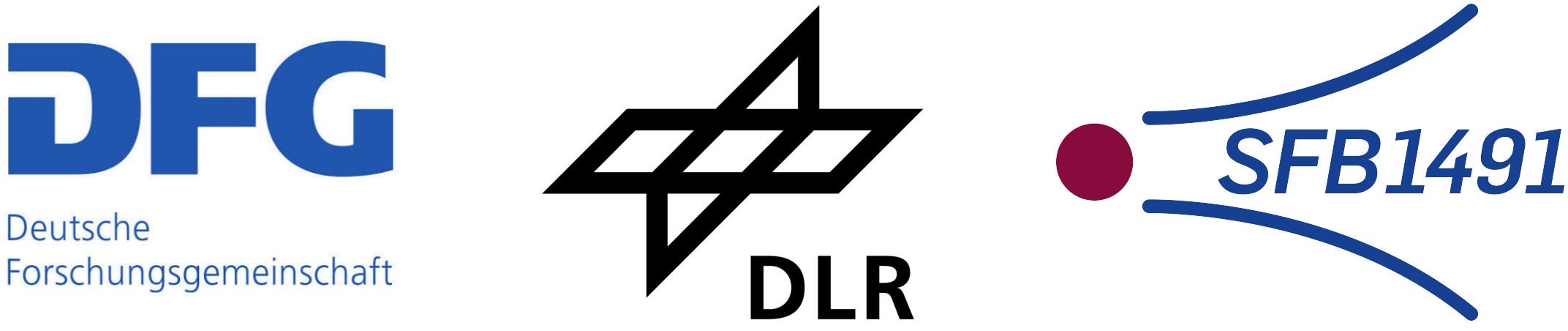


Space Weather Across the Sun–Earth Chain: Our Research

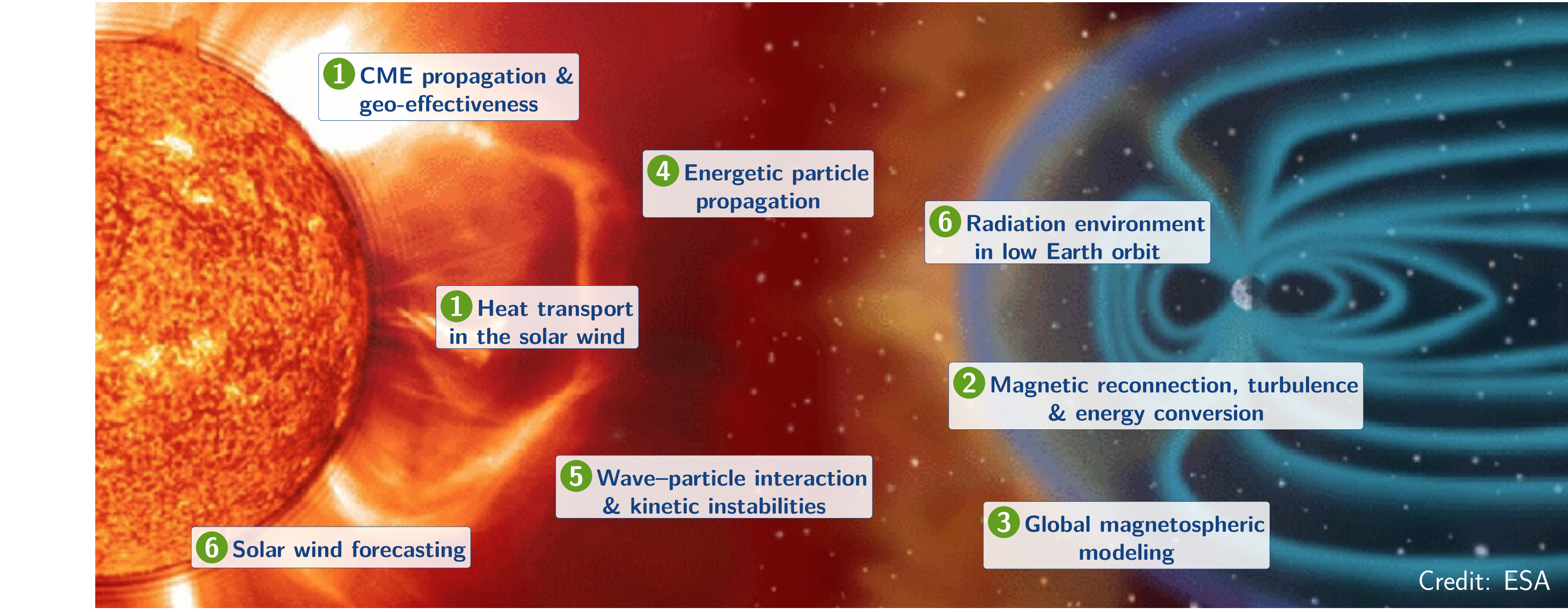
Acknowledgements



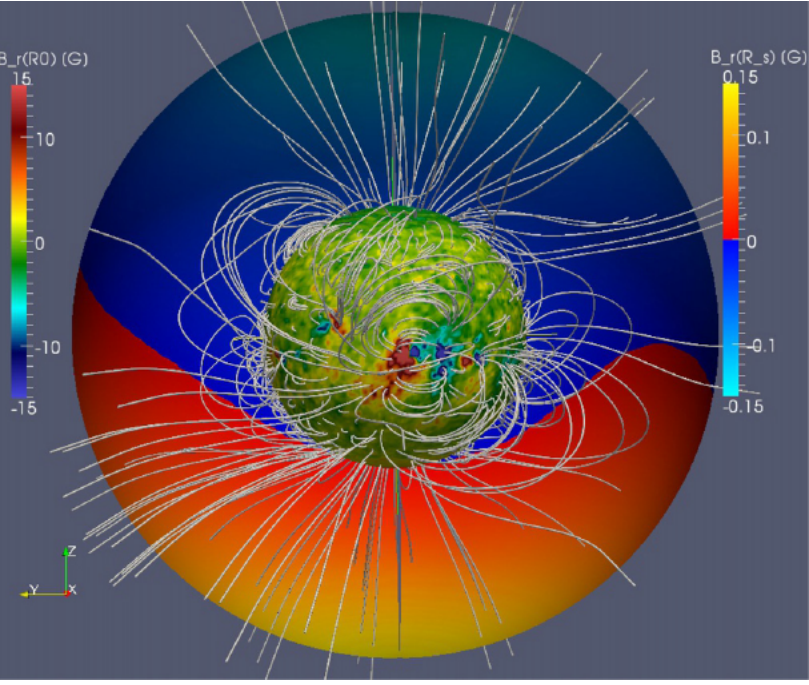
Activities Overview

- Sun and heliosphere:** global heliospheric modeling with CRONOS MHD including co-evolved turbulence transport; CME propagation and geoeffectiveness; interaction between global and kinetic effects in the expanding solar wind, with particular focus on electron and ion-scale instabilities (e.g. González et al. 2023; Micera et al. 2021)
- Energetic particles:** particle transport modeling with CRPropa and Sep2D; shock acceleration; SEP nowcasting
- Geospace:** magnetic reconnection and energy conversion in the terrestrial magnetosphere, with focus on particle heating and acceleration at dipolarization fronts (Aravindakshan et al. 2026); magnetosheath turbulence (Arrò et al. 2023)
- Methods:** explicit and semi-implicit Particle In Cell codes, also with expanding-box capabilities; muphyII framework (Allmann-Rahn et al. 2024); coupled Vlasov, 10-moment and 5-moment methods; MHD coupling in progress; model augmentation with Machine Learning
- Machine Learning for space physics:** clustering and classification of simulated and spacecraft data (e.g. Innocenti et al. 2021; Radenhäuser et al. 2026); ML closures of fluid models; solar wind forecasting with a modified Surya model
- the RUB space weather portal**

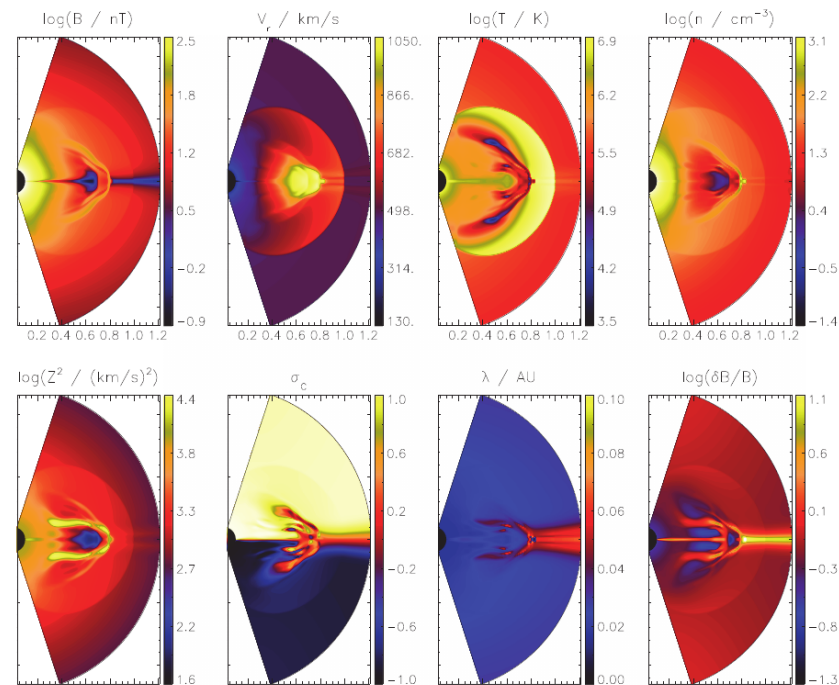
A chance to connect with the broader **European Heliophysics Community** and identify where our work can contribute.



1 The Turbulent Solar Wind: global modeling with CRONOS



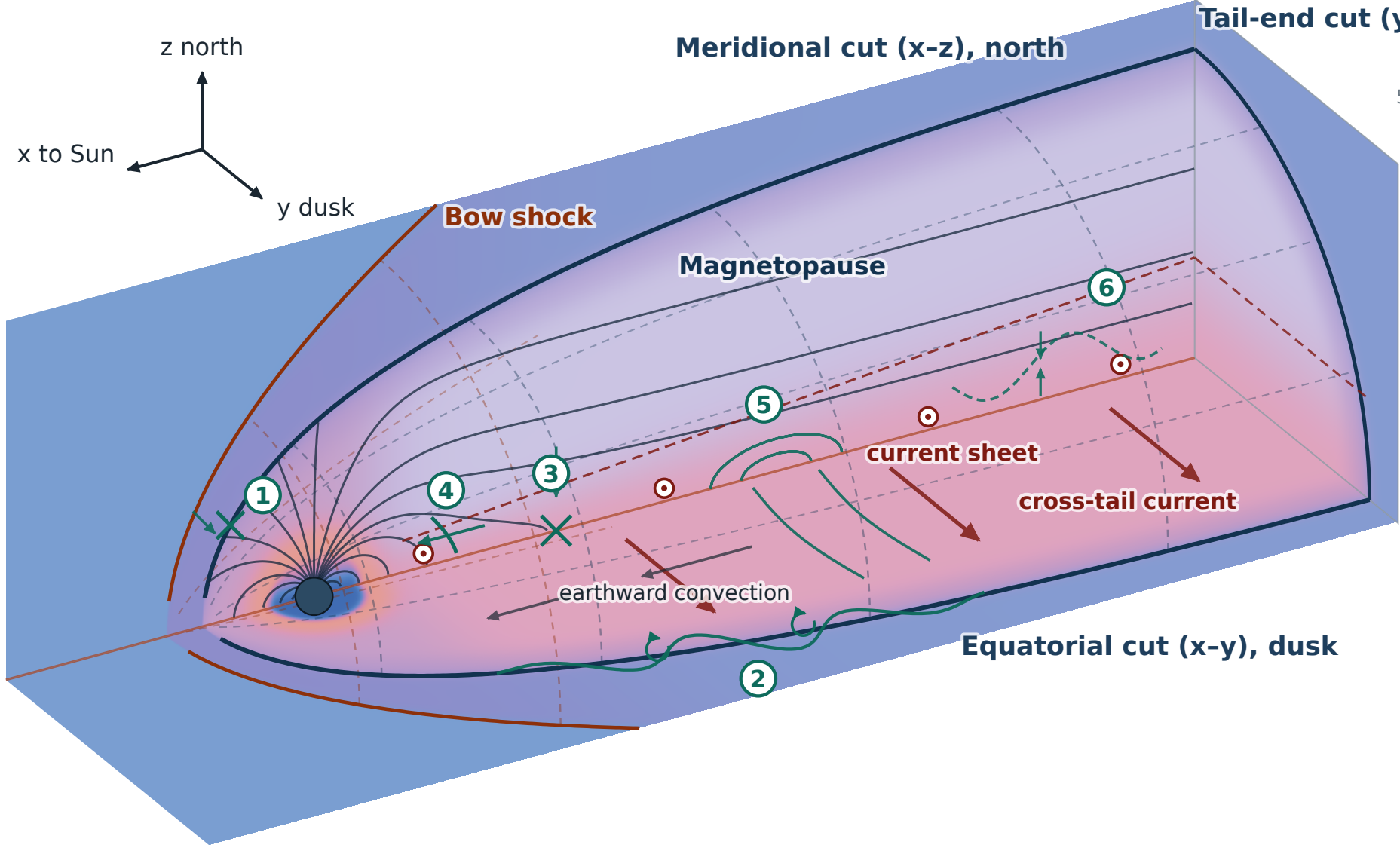
CRONOS inner boundary: coronal field and source surface



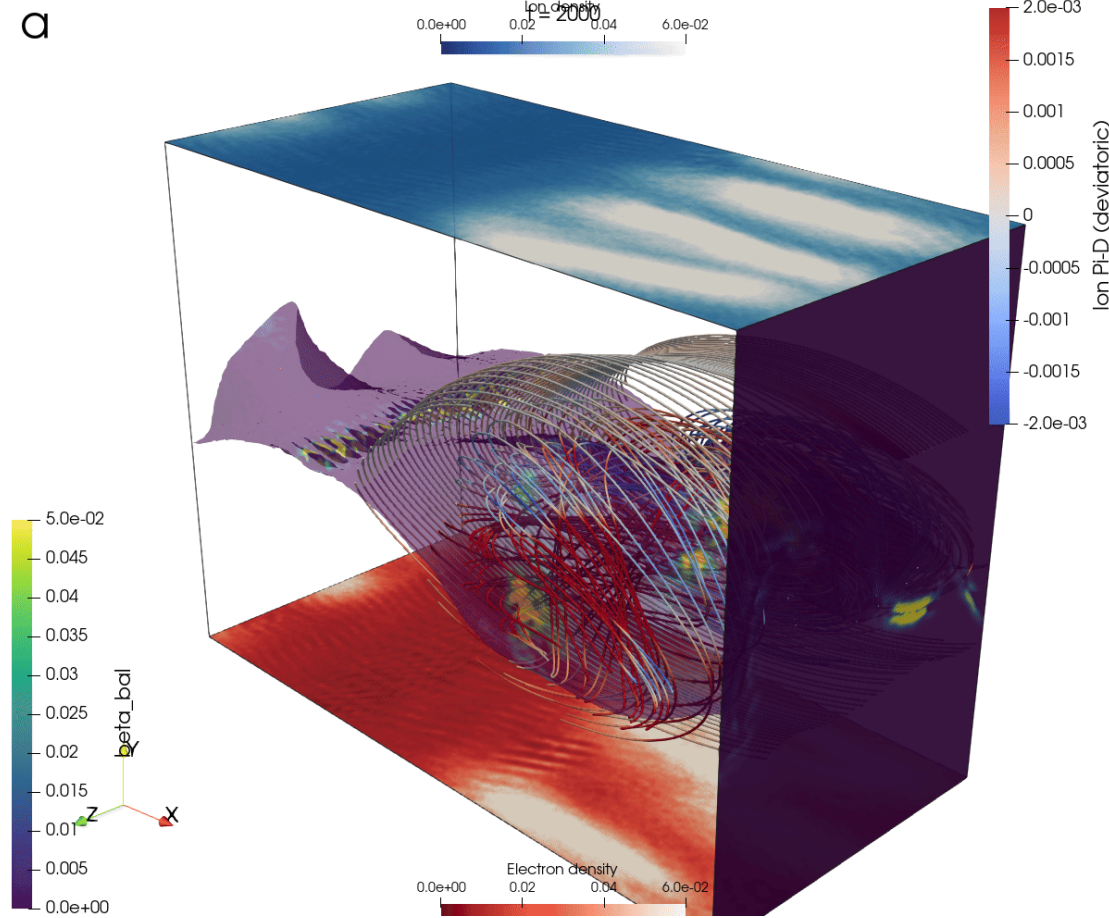
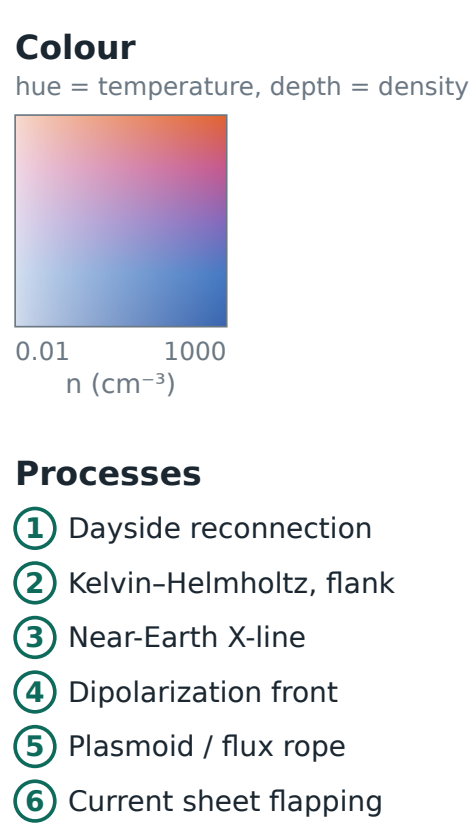
CRONOS to 1.2 AU: background wind and co-evolved turbulence.

- The inner boundary at 0.1 AU, the *heliobase*, is built from GONG magnetograms through a PFSS extrapolation and the WSA relation
- The in-house **CRONOS** MHD code (Kissmann et al. 2018) carries the wind outward, solving turbulence transport equations *alongside* the MHD equations: Elsässer energy Z^2 , cross helicity σ_c and correlation length λ evolve with the flow rather than being prescribed.
- Runs are validated in space and time against STEREO A/B, SOHO and ACE at 1 AU and Ulysses beyond
- Newer turbulence models add intermittency, coherent structures and field-line curvature, which can drive anomalous transport.
- Work in progress: investigation of the role of the turbulent **CME sheath** in driving geomagnetic activity.

2 Particle Heating and Acceleration by Magnetic Reconnection in the Terrestrial Magnetosphere



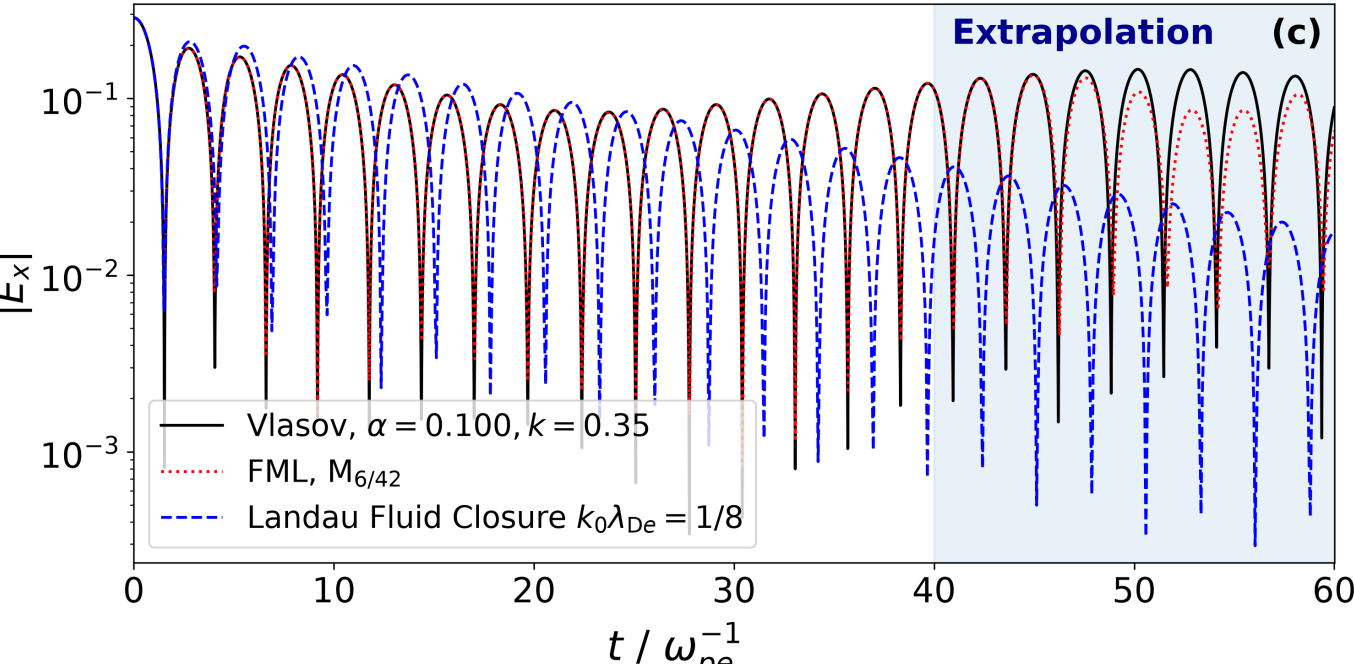
Key dynamical processes in the coupled magnetospheric system.



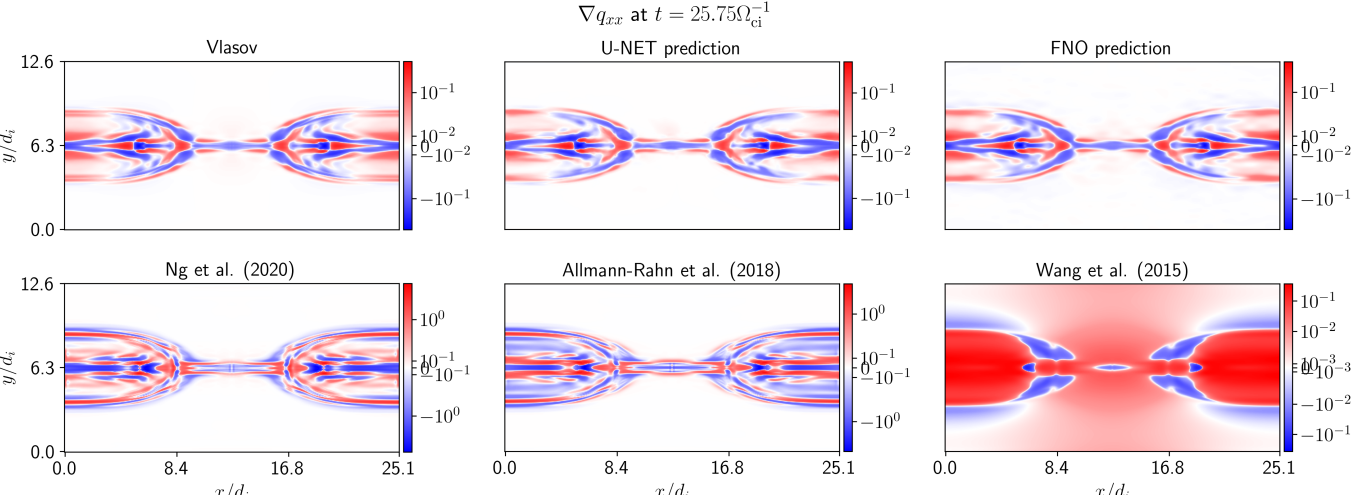
3D fully kinetic simulation of a dipolarization front: magnetic field lines, ion (top) and electron (bottom) density. Traces of ion (ballooning) and electron (LHDI)-scale instabilities are visible in the density

- The magnetosphere responds to events of solar origin as one coupled system: dayside reconnection and the Kelvin–Helmholtz instability mix solar wind and magnetospheric plasmas, and dayside magnetopause reconnection drives magnetotail reconnection
- The leading edges of the resulting fast earthward flows are **dipolarization fronts**: a sharp rise in B with a plasma depletion just ahead. They are efficient particle accelerators and carry much of the energy transported earthward (Aravindakshan et al. 2026)
- Downstream of the bow shock sits the **magnetosheath**, where shocked solar wind becomes strongly turbulent, driving instability and coherent structure formation (Arrò et al. 2023).

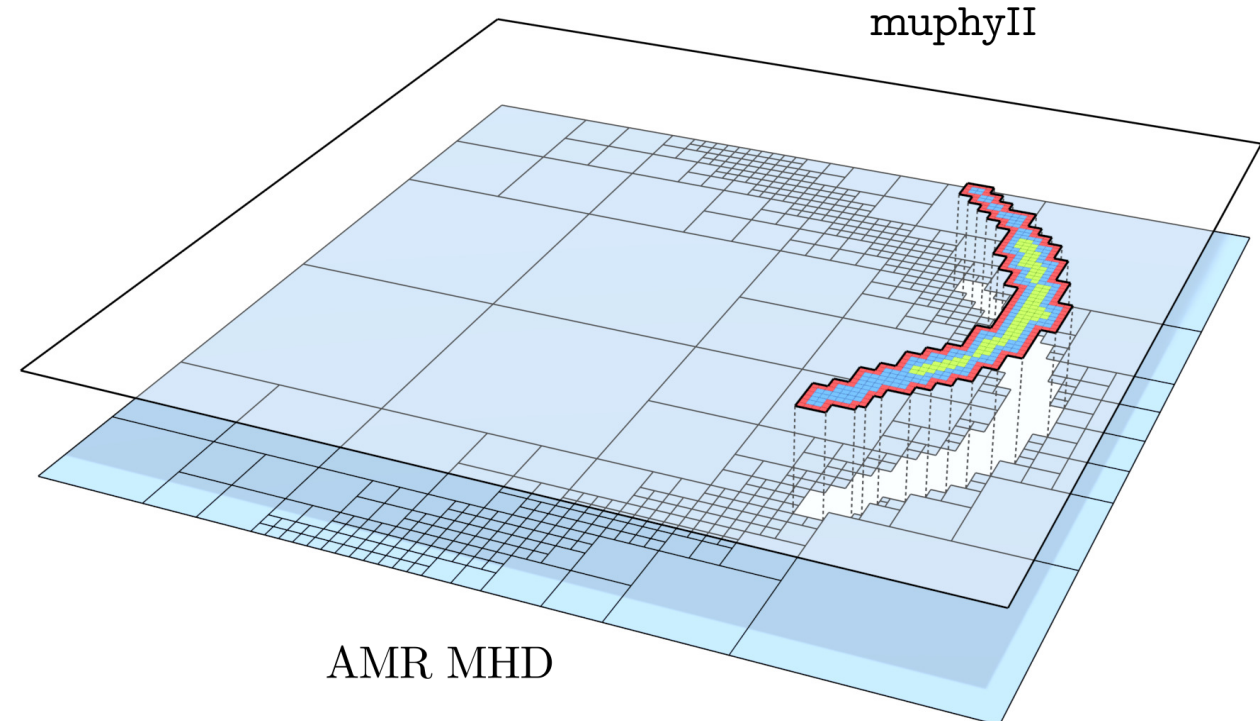
3 Model Hierarchy and Code Coupling with muphyII



FNO and Landau closures for nonlinear Landau damping



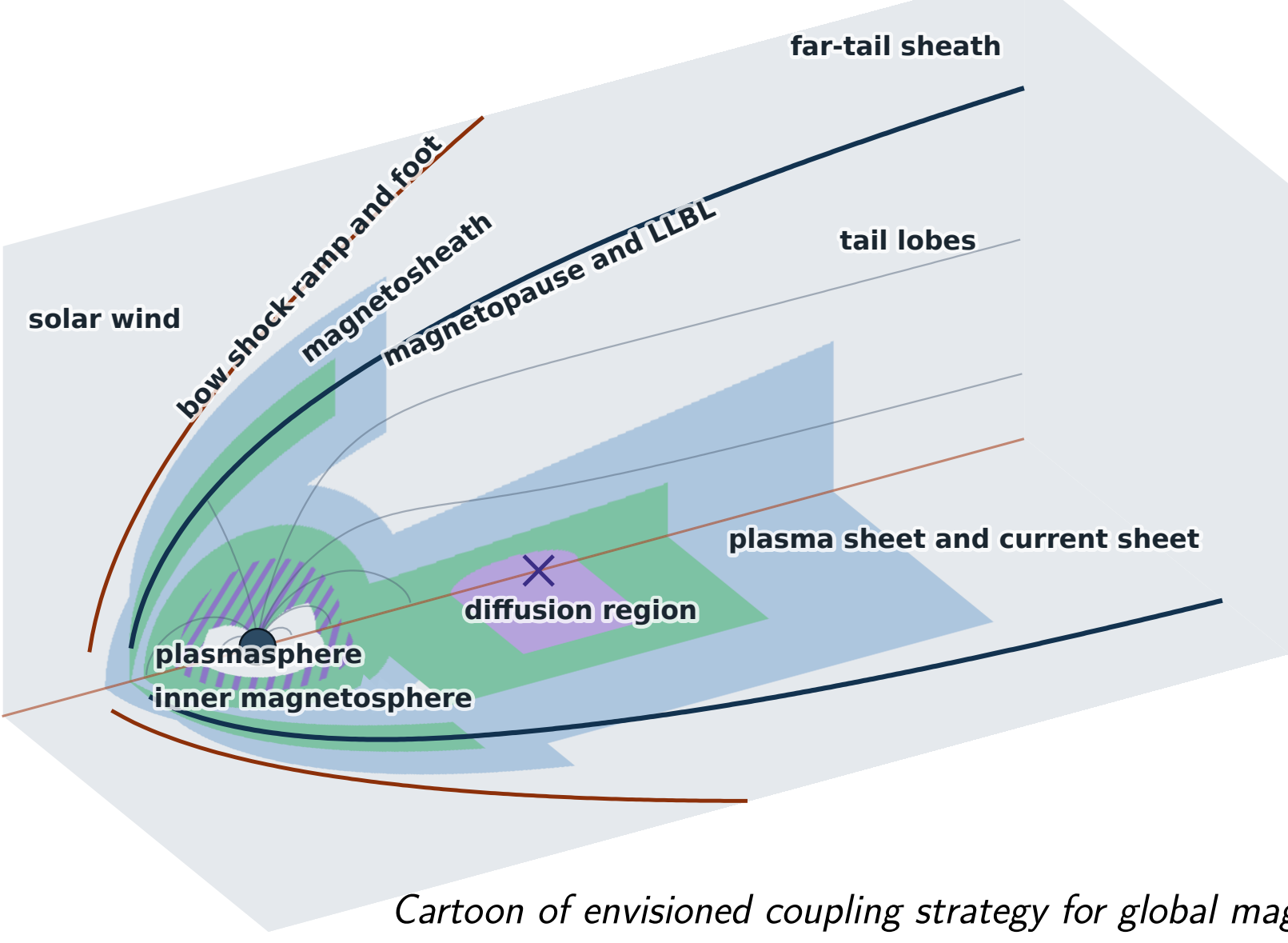
ML and symbolic closures for magnetic reconnection



AMR MHD-muphyII code coupling cartoon for a bow shock

Kinetic simulations of the global magnetosphere are prohibitively expensive for operational approaches. We pursue a **code coupling strategy**

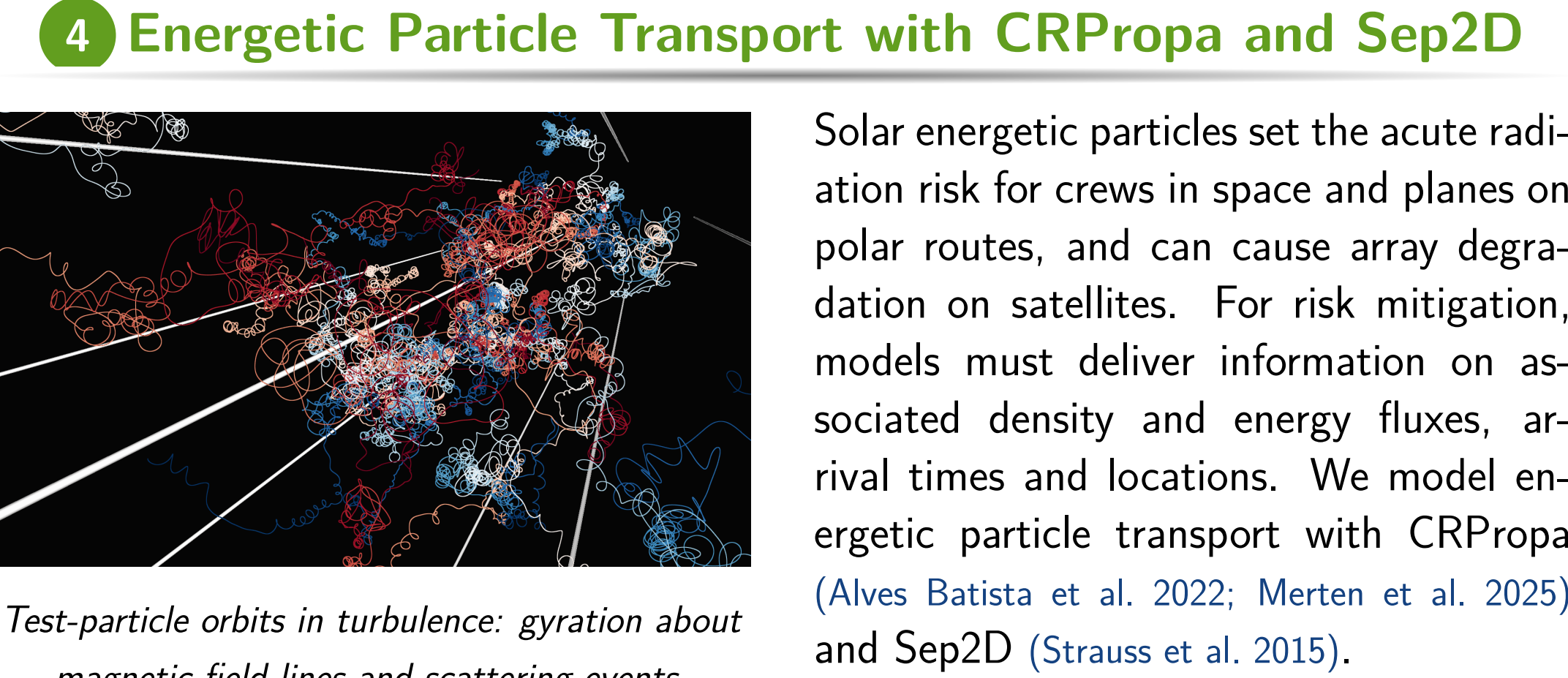
- 5-moment multi-fluid:** evolves density, momentum and scalar pressure per species. Cheap and robust, keeps the two-fluid physics (Hall term, electron inertia) that MHD drops, but assumes isotropic pressure
- 10-moment multi-fluid:** evolves velocity moments up and including the full pressure tensor per species, so anisotropy and agyrotropy survive. Landau-fluid and gradient closures are available (Lautenbach et al. 2026)
- ML closures** learned from kinetic runs are being integrated into the 10-moment description, for a more accurate description of wave/ particle energy transfer (Jeß et al. 2026)
- 10-moment/ 5-moment coupling:** we implement 5-moment and 10-moment patches in muphyII (Allmann-Rahn et al. 2024; Lautenbach et al. 2018), run block-adaptively where needed; asymptotic-preserving MHD/5-moment coupling in progress (Deisenhofer et al. 2026)



Cartoon of envisioned coupling strategy for global magnetospheric simulations

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Test-particle orbits in turbulence: gyration about magnetic field lines and scattering events.

5 Our Codes

Fully kinetic Particle-in-Cell codes

- Explicit: OSIRIS (Fonseca et al. 2002; Schoeffler et al. 2025)
- Semi-implicit, Implicit Moment Method: iPIC3D (Innocenti et al. 2017; Markidis et al. 2010); with expanding-box capabilities: EB-iPIC3D (Innocenti et al. 2019)
- Semi-implicit, Energy Conserving: ECsim (Boella et al. 2026; Lapenta 2017)
- Geometry-preserving: GEMPIC (Kraus et al. 2017)

The muphyII framework (Allmann-Rahn et al. 2024; Lautenbach et al. 2018)

- Vlasov, 10-moment and 5-moment models, coupled in one framework
- MHD coupling in progress

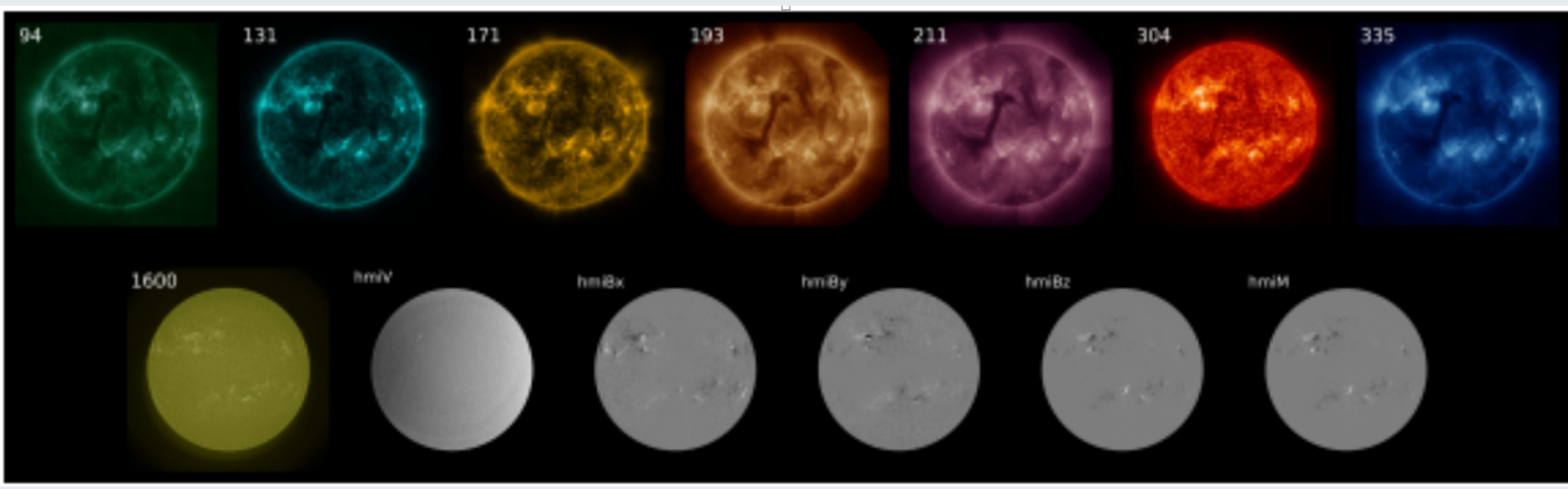
CRONOS (Kissmann et al. 2018)

- Global MHD heliospheric model, data driven
- CME propagation
- (Usmanov et al. 2011) and (Oughton et al. 2011) turbulence models

Particle transport codes

- CRPropa (Alves Batista et al. 2022; Merten et al. 2025): Monte Carlo test-particle code for particles in turbulent magnetic fields; transport models: eqs. of motion (more accurate, more expensive) or stochastic differential eqs. (stat. treatment, cheaper)
- Sep2D (Strauss et al. 2015): SEP transport (van den Berg et al. 2020) prototype model driven by GOES M+ flare data

6 Forecasting, ML and the RUB Portal



The Surya input stack: SDO/AIA passbands with HMI magnetograms.

- we use the Surya model for forecasting of the solar state
- Surya** is a 366M-parameter spatiotemporal transformer trained on full-resolution SDO imagery, predicting future solar states from recent history. We serve it live through our portal.
- we use actual SDO images and Surya-generated solar images for ML prediction of the solar wind at 1 AU
- these forecasts and much more will soon be available in the **RUB Space Weather Portal** live dashboard.

